



Course Prepared by  
IITian & AI Architect

Duration: 7 Months

5 Months Learning + 2 Months Internship

# ADVANCED AI/ML CERTIFICATION PROGRAM

*Learn AI. Build Intelligent Applications. Engineer Real-World AI Systems.*

SKILL IT Education presents a structured, practical career pathway in AI Engineering — taking learners step-by-step from Python foundations and backend development to Machine Learning, Generative AI, LLM application development, Agentic AI, and production-grade AI systems, reinforced with hands-on projects and industry-aligned practices.

## GET IN TOUCH

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## PROGRAM HIGHLIGHTS

- ✓ 7 Core Engineering Modules
- ✓ 5+ Hands-On Capstone Projects
- ✓ LLMs, RAG & Agentic AI
- ✓ Production-Grade Deployment

## YOUR AI ENGINEERING JOURNEY

Python

Backend

ML

GenAI/LLM

Agentic AI

MLOps

AI Solutions

## WHO THIS PROGRAM IS FOR

Freshers, developers and professionals who want to move from writing code to engineering complete, production-grade AI systems.

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# 01 Program Overview

## Build Your AI Engineering Career

### What This Program Covers

This program takes you step-by-step from strong foundations in Python to building intelligent applications with Machine Learning, Generative AI, LLMs, Agentic AI — and finally deploying production-ready AI systems for real business use cases.

### Why This Matters

AI now drives decision-making, automation and product innovation across every industry. Engineers who can design, build and deploy AI systems — not just use AI tools — are in the highest demand across the global technology job market.

#### COURSE DURATION

**5 Months**

#### PROJECT LEARNING

**Integrated**

#### INTERNSHIP

**2 Months**

### PROGRAM ROADMAP

- 1 Python & Technical Foundations
- 2 Python Backend Development with AI
- 3 Machine Learning Engineering
- 4 Generative AI & LLM Application Engineering
- 5 Agentic AI Engineering
- 6 MLOps, LLMOps & AI Platform Engineering
- 7 AI Solutions Engineering

*From Learning to Building. From Projects to Production.  
From Skills to Career.*

## 03 Module 1 — Python & Technical Foundations

### WHAT YOU WILL LEARN

- Python Programming
- Functions & Object-Oriented Programming
- Error Handling
- Virtual Environments
- Git & GitHub Version Control
- Linux / Command Line Interface
- Working with JSON
- HTTP & REST APIs
- Debugging Techniques
- Basic Testing
- Technical Documentation

### TOOLS & TECHNOLOGIES

Python

VS Code

Git

GitHub

Linux

Postman

REST APIs

JSON

#### ✓ LEARNING OUTCOME

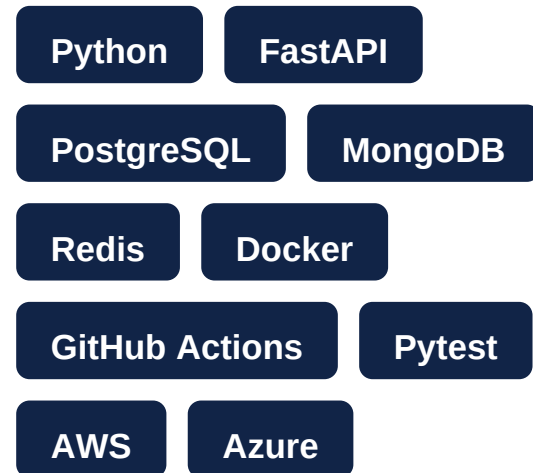
*Students build a strong software foundation for AI development by writing Python applications, working with APIs, managing code with Git/GitHub, debugging applications and applying professional development practices.*

## 04 Module 2 — Python Backend Development with AI

### WHAT YOU WILL LEARN

- Advanced Python & Modular Application Architecture
- FastAPI Development & RESTful API Design
- PostgreSQL & MongoDB Database Fundamentals
- Redis & Background Job Queues
- Authentication, Authorization & JWT / OAuth
- API Security & Role-Based Access Control
- Testing & Test-Driven Development
- Docker Containerization
- CI/CD with GitHub Actions
- Cloud Deployment
- Capstone: Production-Style Backend Service

### TOOLS & TECHNOLOGIES



#### ✓ LEARNING OUTCOME

*Students build robust, secure and scalable backend systems with Python and FastAPI — working with relational and NoSQL databases, implementing authentication and API security, and deploying production-ready services with Docker, CI/CD and the cloud.*

## 05 Module 3 — Machine Learning Engineering

### WHAT YOU WILL LEARN

- Machine Learning Fundamentals
- Supervised Learning
- Unsupervised Learning
- Feature Engineering
- Model Training & Evaluation
- ML Pipelines
- Deep Learning Fundamentals
- Model Serving & Deployment
- Model Monitoring & Retraining

### TOOLS & TECHNOLOGIES



### ML ENGINEERING PIPELINE



#### ✓ LEARNING OUTCOME

*Students learn to develop, evaluate, package, serve and monitor Machine Learning models, and to build practical, production-ready ML services end to end.*

## 06 Module 4 — Generative AI & LLM Application Engineering

### WHAT YOU WILL LEARN

- Generative AI Fundamentals
- Large Language Models
- LLM APIs & Prompt Engineering
- Structured Outputs & Function Calling
- Embeddings & Vector Databases
- Document Ingestion & Chunking
- Retrieval-Augmented Generation (RAG)
- Reranking & RAG Evaluation
- PostgreSQL & pgvector
- MCP Fundamentals

### TOOLS & TECHNOLOGIES

LLM APIs

Hugging Face

LangChain

PostgreSQL

pgvector

Vector DBs

FastAPI

Docker

MCP

### RAG APPLICATION FLOW

Docs

Chunk

Embed

VecDB

Retrieve

LLM

Answer

#### ✓ LEARNING OUTCOME

*Students learn to build LLM-powered applications, integrate AI models with external data and APIs, develop RAG systems, implement vector search and ship practical Generative AI applications.*

## 07 Module 5 — Agentic AI Engineering

### WHAT YOU WILL LEARN

- AI Agent Fundamentals & Architecture
- Tool Calling & Function Calling
- State & Memory Management
- Routing & Agent Workflows
- Model Context Protocol (MCP)
- Multi-Agent Patterns
- Human-in-the-Loop Design
- Retries, Fallbacks & Guardrails
- Agent Evaluation & Observability
- Prompt Injection Defence

### TOOLS & TECHNOLOGIES

Python

LangGraph

MCP

LLM APIs

FastAPI

PostgreSQL

Vector DBs

Docker

### AGENT ARCHITECTURE

User

Agent

Reasoning

MCP/APIs

Action

Evaluate

#### ✓ LEARNING OUTCOME

*Students design, build and evaluate AI agents that call tools, maintain state, execute multi-step workflows, and operate safely and reliably in production.*

## 08 Module 6 — MLOps, LLMOps & AI Platform Engineering

### WHAT YOU WILL LEARN

- Experiment Tracking & Model Registry
- Model & Prompt Versioning
- AI & RAG Evaluation
- Regression Testing
- CI/CD for AI Systems
- Tracing & Observability
- Cloud IAM & Secrets Management
- Scaling & Rollbacks
- Cost & Latency Management
- AI Security & Reliability

### TOOLS & TECHNOLOGIES

MLflow

Git

GitHub Actions

Docker

Cloud Platform

PostgreSQL / pgvector

CI/CD

Monitoring Tools

### AI PLATFORM LIFECYCLE

Develop

Test

Evaluate

Deploy

Monitor

Improve

#### ✓ LEARNING OUTCOME

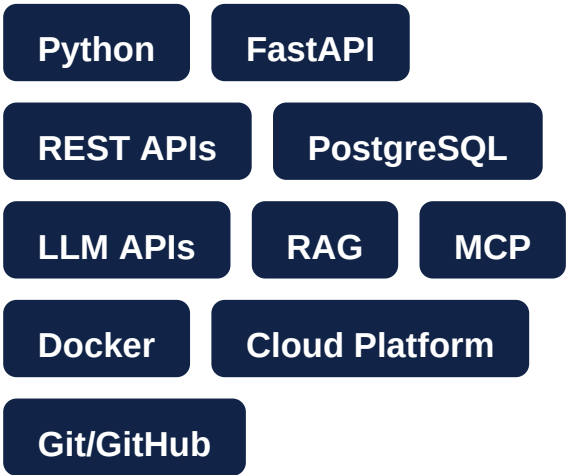
*Students learn to operationalize AI systems end-to-end — tracking experiments, versioning models and prompts, automating testing and deployment, and monitoring production AI for cost, latency, security and reliability.*

# 09 Module 7 — AI Solutions Engineering

## WHAT YOU WILL LEARN

- Client Discovery & Requirement Gathering
- Workflow Mapping
- AI Solution Architecture
- Data & System Integration
- API Integration
- Security & PII Handling
- Solution Design & Prototyping
- UAT & Deployment
- Cost Estimation & Technical Documentation
- Architecture Diagrams & Client Demonstrations

## TOOLS & TECHNOLOGIES



## SOLUTION DEVELOPMENT JOURNEY



### ✓ LEARNING OUTCOME

*Students learn to gather requirements, design AI solution architectures, integrate systems and APIs, and prototype and deploy client-ready AI solutions with proper documentation and demonstration.*

# 10 AI Engineering Technology Stack

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*The complete toolset used across every module of the program*

## **Programming & Development**

- Python
- Git & GitHub
- Linux
- VS Code

## **Backend & APIs**

- FastAPI
- REST APIs
- PostgreSQL
- MongoDB

## **Data & Retrieval**

- pgvector
- Vector Databases
- Redis

## **Machine Learning**

- Scikit-learn
- PyTorch
- MLflow

## **Generative AI**

- LLM APIs
- Hugging Face
- LangChain
- RAG

## **Agent Engineering**

- LangGraph
- MCP
- Tool Calling
- Tracing

## **DevOps & Testing**

- Docker
- GitHub Actions
- Pytest

## **Cloud Platforms**

- AWS
- Azure

# 11 Project-Based Learning

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*Build what you learn — every module is reinforced with hands-on work*

## 01 Python AI Application

Build Python applications using APIs, data processing pipelines and automation scripts.

## 02 Backend API Service

Build and deploy a secure, production-style REST API with FastAPI, PostgreSQL/MongoDB, authentication and Docker.

## 03 Machine Learning Service

Train, evaluate and deploy Machine Learning models as production-ready services.

## 04 Generative AI / RAG Application

Build LLM-powered apps with RAG, vector search and document understanding.

## 05 Agentic AI Application

Build intelligent agents that use tools, make decisions and execute multi-step tasks.

### FINAL CAPSTONE

#### End-to-End AI Solution

Design and build a complete AI solution from requirements to deployment and live demonstration, bringing together Python, backend engineering, Machine Learning, Generative AI and Agentic AI.

## 12 Learning Outcomes

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*What you will be able to build by the end of the program*

- ✓ Develop AI applications using Python
- ✓ Build and evaluate Machine Learning models
- ✓ Build LLM-powered applications
- ✓ Work with embeddings and vector databases
- ✓ Integrate external tools and APIs
- ✓ Deploy AI applications to production
- ✓ Apply AI security and reliability practices
- ✓ Build, present and demonstrate real AI projects
- ✓ Build and deploy secure backend APIs & services
- ✓ Develop deep learning solutions
- ✓ Develop Retrieval-Augmented Generation (RAG) systems
- ✓ Build and orchestrate AI agents
- ✓ Work with LangGraph and MCP
- ✓ Evaluate and monitor AI systems
- ✓ Design end-to-end AI solution architectures

# 13 Career Opportunities

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*Build your career in AI*

## **Backend & Software Engineering**

- Backend Developer
- API Developer
- Python Developer
- Software Engineer

## **AI & Machine Learning**

- AI Engineer
- Machine Learning Engineer
- MLOps Engineer
- Machine Learning Developer

## **Generative AI**

- Generative AI Engineer
- GenAI Application Engineer
- LLM Application Developer
- AI Application Developer

## **Agentic AI**

- Agentic AI Engineer
- AI Agent Developer
- AI Automation Engineer
- AI Workflow Developer

## **AI Production & Platform**

- MLOps Engineer
- LLMOps Engineer
- AI Platform Engineer
- AI Infrastructure Engineer

## **AI Solutions**

- AI Solutions Engineer
- AI Integration Engineer
- AI Implementation Engineer
- AI Consultant

Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.

## 14 Technical Assessment & Industry Certifications

*Measure your skills. Improve your performance.*

### Weekly Assessments

- Regular technical assessments to measure understanding of recently completed topics.

### Module-End Assessments

- Comprehensive technical evaluation after completion of each major module.

### Practical / Coding Assessments

- Students demonstrate their ability to implement concepts through coding and real-world tasks.

### Project Assessments

- Evaluate project architecture, implementation, testing, documentation and presentation.

## CERTIFICATION PATHWAYS ALIGNED WITH COURSE CONTENT

Microsoft Azure AI Engineer

AWS ML Engineer / Cloud Practitioner

Google Cloud Professional ML Engineer

NVIDIA Generative AI & LLMs

Databricks ML / GenAI Engineer

Oracle Cloud Infrastructure AI

IBM AI / Generative AI Engineering

TensorFlow & Hugging Face Credentials

### Certification Notice

The curriculum is aligned to help students prepare for these external certifications. Registration, fees and eligibility for any external exam are managed directly by the respective certifying body and may change over time. SKILL IT Education awards its own course-completion certificate upon successful completion of the program.

## 15 Internship, Industry Exposure & Placement Support

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After successful course completion and applicable project evaluations, eligible students can progress toward internship and practical industry exposure opportunities — gaining hands-on experience across AI Application Development, Machine Learning, Generative AI/LLM, RAG, Agentic AI, AI Automation, API Integration and AI Deployment, subject to availability and selection criteria.

### 01 Resume Preparation

Guided support to build a strong, role-ready resume.

### 02 Portfolio Guidance

LinkedIn & GitHub portfolio guidance to showcase your projects.

### 03 Project Presentation

Preparation to confidently present your capstone projects.

### 04 Interview Preparation

Technical interview preparation with mock interviews & coding practice.

### 05 HR & Communication Prep

HR interview and communication preparation.

### 06 Job Search Support

Job search & interview opportunity sharing with feedback and guidance.

### Complete Career Transition Support

Skill IT Education's Placement Assistance Team works closely with every learner — guiding them through the full Career Readiness Journey, from building skills and projects to assessments, portfolio, mock interviews, internship and final placement.



## WHY CHOOSE SKILL IT EDUCATION

# DON'T JUST LEARN AI. LEARN TO ENGINEER IT.

*Learn. Build. Deploy. Evaluate. Demonstrate.*

SKILL IT Education builds AI engineers, not just AI users. Through an industry-driven curriculum, project-based learning, real-world tools and hands-on capstone work, every student graduates with a portfolio of production-style AI systems, verified skills and the confidence to walk into technical interviews and deliver from day one — in Machine Learning, Generative AI, Agentic AI and beyond.

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*Let's build the future with AI.*

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